ORDINARY DIFFERENTIAL EQUATIONS • UNIT 2.01

# C Linear First-Order Equations & Integrating Factors

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

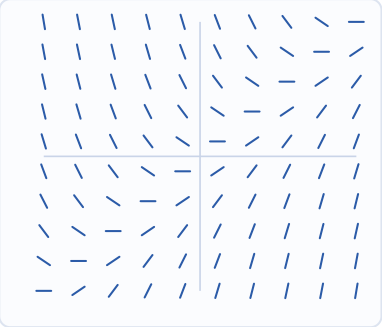
## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 A | Q2 D | Q3 D | Q4 A | Q5 A |
| Q6 A | Q7 D | Q8 A | Q9 A | Q10 B |
| Q11 D | Q12 C | Q13 C | Q14 D | Q15 B |
| Q16 A | Q17 D | Q18 B | Q19 B | Q20 B |
| Q21 A | Q22 C | Q23 A | Q24 D | Q25 C |

## Linear First-Order Equations & Integrating Factors Practice Exercises

**1** MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots $\lambda_1, \lambda_2$. Determine the trace sum $\lambda_1 + \lambda_2$.



- **A** 39 ✓
- **B** 24
- **C** 42
- **D** 53

**WORKED SOLUTION & VERIFICATION**

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is $39$.

**2** MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

B $e^2 - 1$

C $3 - 2e^{-2}$

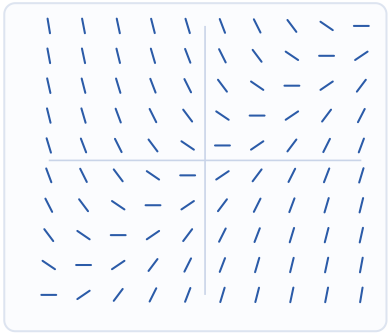
D $3 - 2e^{-1}$ ✓

**WORKED SOLUTION & VERIFICATION**

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$.
Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$.
Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$.
The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$.
Integrating both sides with respect to $x$: $ye^{2x} = \int 6e^{2x}dx$.
$ye^{2x} = 6 \cdot \frac{1}{2}e^{2x} + C$.
$ye^{2x} = 3e^{2x} + C$.
Solving for $y$: $y = 3 + Ce^{-2x}$.
Using the initial condition $y(0) = 1$ to find $C$: $1 = 3 + Ce^{-2(0)}$.
$1 = 3 + C$.
$C = 1 - 3 = -2$.
So, the particular solution is $y(x) = 3 - 2e^{-2x}$.
Now, we need to find $y(1)$: $y(1) = 3 - 2e^{-2(1)} = 3 - 2e^{-2}$.
This matches choice C.

**3** MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D **1.220** ✓

**WORKED SOLUTION & VERIFICATION**

Using Euler's method $y_{n+1} = y_n + h(x_n + y_n)$ with $x_0 = 0, y_0 = 1, h = 0.1$:
1. $y_1 = 1 + 0.1(0 + 1) = 1.100$ at $x_1 = 0.1$.
2. $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.220$ at $x_2 = 0.2$.

**4** MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots $r_1$ and $r_2$, find the value of $r_1 + r_2$.

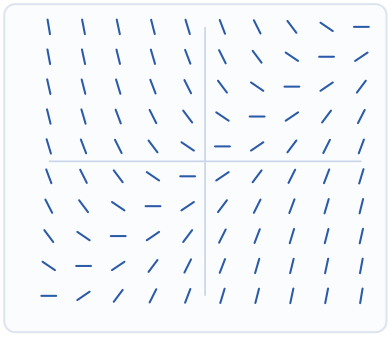
- **A** 5 ✓
- **B** -5
- **C** 6
- **D** -6

**WORKED SOLUTION & VERIFICATION**

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be $r_1$ and $r_2$. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation $r^2 - 5r + 6 = 0$, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $r_1 + r_2 = -\frac{-5}{1} = 5$.

**5** MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$ .



- **A** 7 ✓
- **B** 6
- **C** 8
- **D** 5

**WORKED SOLUTION & VERIFICATION**

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular solution, we have $y(2) = (2)^2 + 3 = 4 + 3 = 7$.

**6** MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients $c_n$.

- **A** $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓
- **B** $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$
- **C** $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$
- **D** $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

**WORKED SOLUTION & VERIFICATION**

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to $x$, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of $x$ to be the same. Let $k = n-2$ in the first series, so $n = k+2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n+1$ in the second series, so $n = k-1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all $x$, the coefficients of each power of $x$ must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing $k$ with $n$, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

**7** MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots $\lambda_1, \lambda_2$. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** 24
- **D** 40 ✓

**WORKED SOLUTION & VERIFICATION**

From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is $40$.

**8** MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

- **A** 170 ✓
- **B** 151
- **C** 173
- **D** 187

**WORKED SOLUTION & VERIFICATION**

Substituting $s = 20$ gives $\frac{170}{20-19} = 170$.

**9** MCQ

The Laplace transform of $f(x) = 95e^{17x}$ is $g(s) = \dfrac{95}{s-17}$ for $s > 17$. Determine the evaluated transform $g(18)$.

- **A** 95 ✓
- **B** 98
- **C** 78
- **D** 100

**WORKED SOLUTION & VERIFICATION**

Substituting $s = 18$ gives $\frac{95}{18-17} = 95$.

**10** MCQ

Consider the first-order linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$. If the integrating factor for this equation is given by $e^{x^2}$ and $Q(x) = 2xe^{x^2}$, find the value of the constant $C$ in the general solution if $y(0) = 3$. The general solution is of the form $y = e^{-x^2}\left(\int Q(x)e^{-x^2}\,dx + C\right)$.

- **A** 1
- **B** 3 ✓
- **C** 4
- **D** 6

**WORKED SOLUTION & VERIFICATION**

The given differential equation is in the form $\frac{dy}{dx} + P(x)y = Q(x)$. The integrating factor (IF) is given by $e^{\int P(x)dx}$. We are told that the integrating factor is $e^{x^2}$. Therefore, $\int P(x)dx = x^2$, which implies $P(x) = 2x$.

The general solution to a first-order linear differential equation is given by $y \cdot \text{IF} = \int Q(x) \cdot \text{IF}\,dx + C$.

Substituting the IF and $Q(x)$, we get:

$ye^{x^2} = \int (2xe^{x^2})e^{-x^2}\,dx + C$

$ye^{x^2} = \int 2x\,dx + C$

$ye^{x^2} = x^2 + C$

Now, we need to find the value of $C$ using the initial condition $y(0) = 3$.

Substitute $x = 0$ and $y = 3$ into the general solution:

$3e^{0^2} = 0^2 + C$

$3 \cdot 1 = 0 + C$

$C = 3.$

Therefore, the value of the constant $C$ is 3.

**11** MCQ

The separable initial value problem $\dfrac{dU}{dv} = 28v$ satisfies $U(0) = 9$. Determine the solution value $U(16)$.

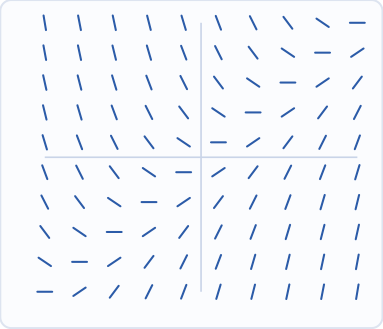
- **A** 3584
- **B** 3607
- **C** 3596
- **D** 3593 ✓

**WORKED SOLUTION & VERIFICATION**

Integrating gives $U(v) = 14v^2 + 9$, so $U(16) = 3593$.

**12** MCQ

The separable initial value problem $\frac{dP}{dr} = 34r$ satisfies $P(0) = 8$. Determine the solution value $P(7)$.



- A 858
- B 844
- C **841** ✓
- D 833

**WORKED SOLUTION & VERIFICATION**

Integrating gives $P(r) = 17r^2 + 8$, so $P(7) = 841$.

**13** MCQ

The Laplace transform of $R(m) = 48e^{20m}$ is $S(s) = \frac{48}{s - 20}$ for $s > 20$. Determine the evaluated transform $S(21)$.

- A 28
- B 51
- C **48** ✓
- D 56

**WORKED SOLUTION & VERIFICATION**

Substituting $s = 21$ gives $\frac{48}{21-20} = 48$.

**14** MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots $\lambda_1, \lambda_2$. Determine the trace sum $\lambda_1 + \lambda_2$.

- A 45
- B 50
- C 23
- D **42** ✓

**WORKED SOLUTION & VERIFICATION**

From $\lambda^2 - 42\lambda + 272 = 0$, the sum of roots is $42$.

**15** MCQ

The separable initial value problem $\frac{dC}{dq} = 12q$ satisfies $C(0) = 14$. Determine the solution value $C(18)$.

A 1944

B **1958** ✓

C 1961

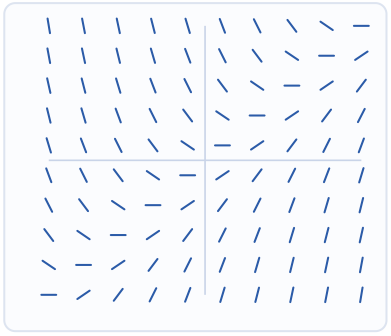
D 1964

**WORKED SOLUTION & VERIFICATION**

Integrating gives $C(q) = 6q^2 + 14$, so $C(18) = 1958$.

**16** MCQ

The Laplace transform of $E(p) = 100e^{24p}$ is $L(s) = \frac{100}{s-24}$ for $s > 24$. Determine the evaluated transform $L(25)$.



A **100** ✓

B 110

C 76

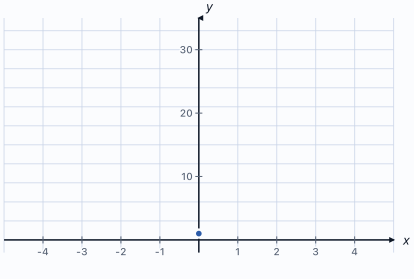
D 103

**WORKED SOLUTION & VERIFICATION**

Substituting $s = 25$ gives $\frac{100}{25-24} = 100$.

**17** MCQ

Solve $y' + 3y = e^{2x}$ with $y(0) = 1$. What is the value of $y(1)$?




A $\frac{1}{5}e^2 + \frac{1}{5}e^{-3}$

B $\frac{1}{2}e^2 + \frac{1}{2}e^{-3}$

C $e^2 + 4e^{-3}$

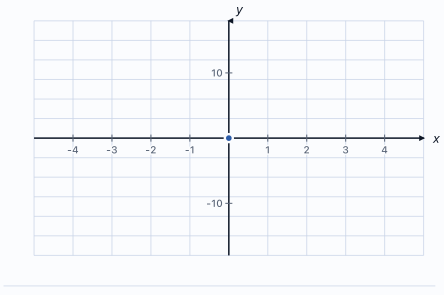
D $\frac{1}{5}e^2 + \frac{4}{5}e^{-3}$ ✓

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int 3\,dx} = e^{3x}$. Multiplying gives $(ye^{3x})' = e^{5x}$. Integrating yields $ye^{3x} = \frac{1}{5}e^{5x} + C \implies y(x) = \frac{1}{5}e^{2x} + Ce^{-3x}$. Using $y(0) = 1$ yields $1 = \frac{1}{5} + C \implies C = \frac{4}{5}$. Thus $y(1) = \frac{1}{5}e^2 + \frac{4}{5}e^{-3}$.

**18** MCQ

Solve $y' + \frac{1}{x}y = 4x$ for $x > 0$ with $y(1) = 3$. What is $y(2)$?




A $\frac{19}{3}$

B $\frac{37}{6}$ ✓

C $6$

D $\frac{35}{6}$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int \frac{1}{x}\,dx} = x$. Multiplying gives $(xy)' = 4x^2$. Integrating yields $xy = \frac{4}{3}x^3 + C \implies y = \frac{4}{3}x^2 + \frac{C}{x}$. Using $y(1) = 3$ gives $3 = \frac{4}{3} + C \implies C = \frac{5}{3}$. At $x = 2$, $y(2) = \frac{4}{3}(4) + \frac{5}{6} = \frac{32}{6} + \frac{5}{6} = \frac{37}{6}$.

**19** MCQ

Solve the initial value problem $y' - 2y = 8$ with $y(0) = 0$. What is $y(\ln 2)$?

A $4$

B $12$ ✓

C $8$

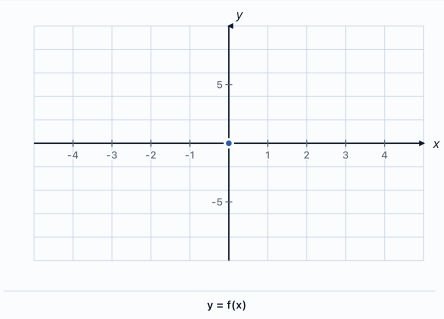
D $16$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int -2\,dx} = e^{-2x}$. Multiplying gives $(ye^{-2x})' = 8e^{-2x}$. Integrating gives $ye^{-2x} = -4e^{-2x} + C \implies y(x) = -4 + Ce^{2x}$. Using $y(0) = 0$ yields $C = 4$, so $y(x) = 4e^{2x} - 4$. At $x = \ln 2$, $e^{2\ln 2} = 4$, so $y(\ln 2) = 4(4) - 4 = 12$.

**20** MCQ

Solve $y' + y = 2x$ with $y(0) = 1$. What is the value of $y(1)$?




A $2 - 3e^{-1}$

B $3e^{-1}$ ✓

C $2 + 3e^{-1}$

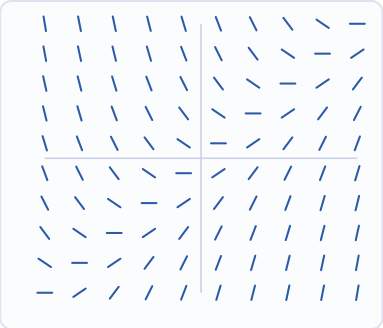
D $e^{-1}$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int 1\,dx} = e^{x}$. Multiplying gives $(ye^{x})' = 2xe^{x}$. Integrating by parts yields $ye^{x} = 2(x-1)e^{x} + C \implies y(x) = 2x - 2 + Ce^{-x}$. Using $y(0) = 1$ gives $1 = -2 + C \implies C = 3$. Thus $y(1) = 2(1) - 2 + 3e^{-1} = 3e^{-1}$.

**21** MCQ

Given $y' + 4y = 12$ with $y(0) = 7$, what is $\lim_{x\to\infty} y(x)$?



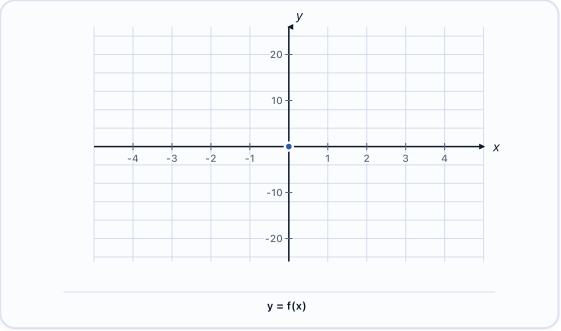
- **A** $3$ ✓
- **B** $12$
- **C** $7$
- **D** $0$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int 4\,dx} = e^{4x}$. Multiplying gives $(ye^{4x})' = 12e^{4x} \implies ye^{4x} = 3e^{4x} + C \implies y(x) = 3 + Ce^{-4x}$. Using $y(0) = 7$ gives $C = 4$, so $y(x) = 3 + 4e^{-4x}$. As $x \to \infty$, $e^{-4x} \to 0$, so $\lim_{x\to\infty} y(x) = 3$.

**22** MCQ

Solve $xy' + 2y = 6x$ for $x > 0$, subject to $y(1) = 4$. What is $y(2)$?



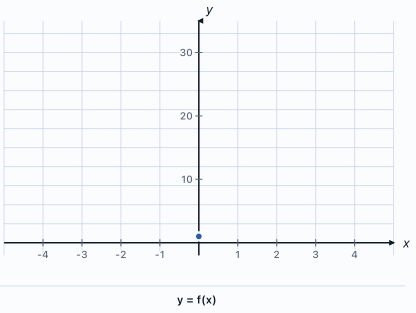

- **A** $\frac{7}{2}$
- **B** $4$
- **C** $\frac{9}{2}$ ✓
- **D** $5$

**WORKED SOLUTION & VERIFICATION**

Rewriting in standard form gives $y' + \frac{2}{x}y = 6$. Integrating factor: $\mu = e^{\int \frac{2}{x}\,dx} = x^2$. Multiplying gives $(x^2y)' = 6x^2 \implies x^2y = 2x^3 + C \implies y = 2x + \frac{C}{x^2}$. Using $y(1) = 4$ yields $4 = 2 + C \implies C = 2$. At $x = 2$, $y(2) = 2(2) + \frac{2}{4} = 4 + \frac{1}{2} = \frac{9}{2}$.

**23** MCQ

Solve $y' - y = e^x$ with $y(0) = 5$. What is the value of $y(1)$?




**A** $6e$ ✓

**B** $7e$

**C** $6e - 1$

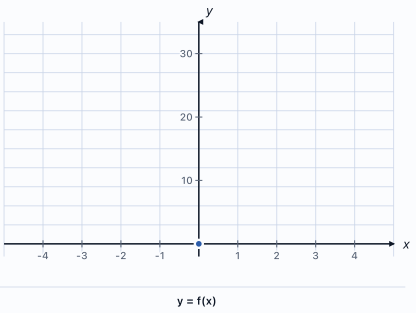
**D** $5e$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int -1\,dx} = e^{-x}$. Multiplying gives $(ye^{-x})' = 1$. Integrating yields $ye^{-x} = x + C \implies y(x) = (x + C)e^x$. Using $y(0) = 5$ gives $C = 5$. Thus $y(x) = (x + 5)e^x$, and $y(1) = (1 + 5)e = 6e$.

**24** MCQ

Solve $y' + 3x^2y = 6x^2$ with $y(0) = 5$. What is the value of $y(1)$?




**A** $2 + 5e^{-1}$

**B** $3 + 2e^{-1}$

**C** $2 - 3e^{-1}$

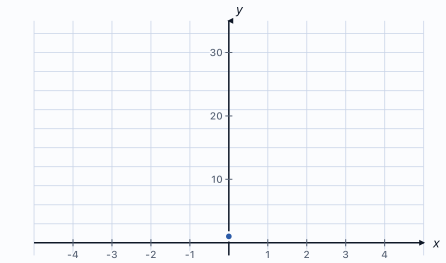
**D** $2 + 3e^{-1}$ ✓

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int 3x^2\,dx} = e^{x^3}$. Multiplying gives $(ye^{x^3})' = 6x^2e^{x^3}$. Integrating yields $ye^{x^3} = 2e^{x^3} + C \implies y(x) = 2 + Ce^{-x^3}$. Applying $y(0) = 5$ gives $5 = 2 + C \implies C = 3$. Thus $y(1) = 2 + 3e^{-1}$.

**25** MCQ

Solve $y' + 2y = e^{-2x}$ with $y(0) = 3$. What is $y(2)$?




A $3e^{-4}$

B $2e^{-4}$

C $5e^{-4}$ ✓

D $6e^{-4}$

**WORKED SOLUTION & VERIFICATION**

$\mu = e^{\int 2\,dx} = e^{2x}$. Multiplying gives $(ye^{2x})' = 1$. Integrating gives $ye^{2x} = x + C \implies y(x) = (x + C)e^{-2x}$. Using $y(0) = 3$ gives $C = 3$. Thus $y(x) = (x + 3)e^{-2x}$, and $y(2) = (2 + 3)e^{-4} = 5e^{-4}$.